Total Probability Theorem:

$$P(A) = P(A \cap B) + P(A \cap B^c)$$
$$= P(A|B)P(B) + P(A|B^c)P(B^c)$$

Generalized total probability theorem:

$B_1, B_2, \ldots, B_n$ partition S

$A \cap B_1, A \cap B_2, \ldots, A \cap B_n$ " A

$$\boxed{P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \ldots + P(A|B_n)P(B_n)}$$

Ex. We have 2 urns

Urn 1 : 9 W Balls & 1 R Ball

" 2 : 1 W ball & 4 R Balls

Experiment (i) toss a fair coin

if heads, choose urn 1

if tails, " " 2

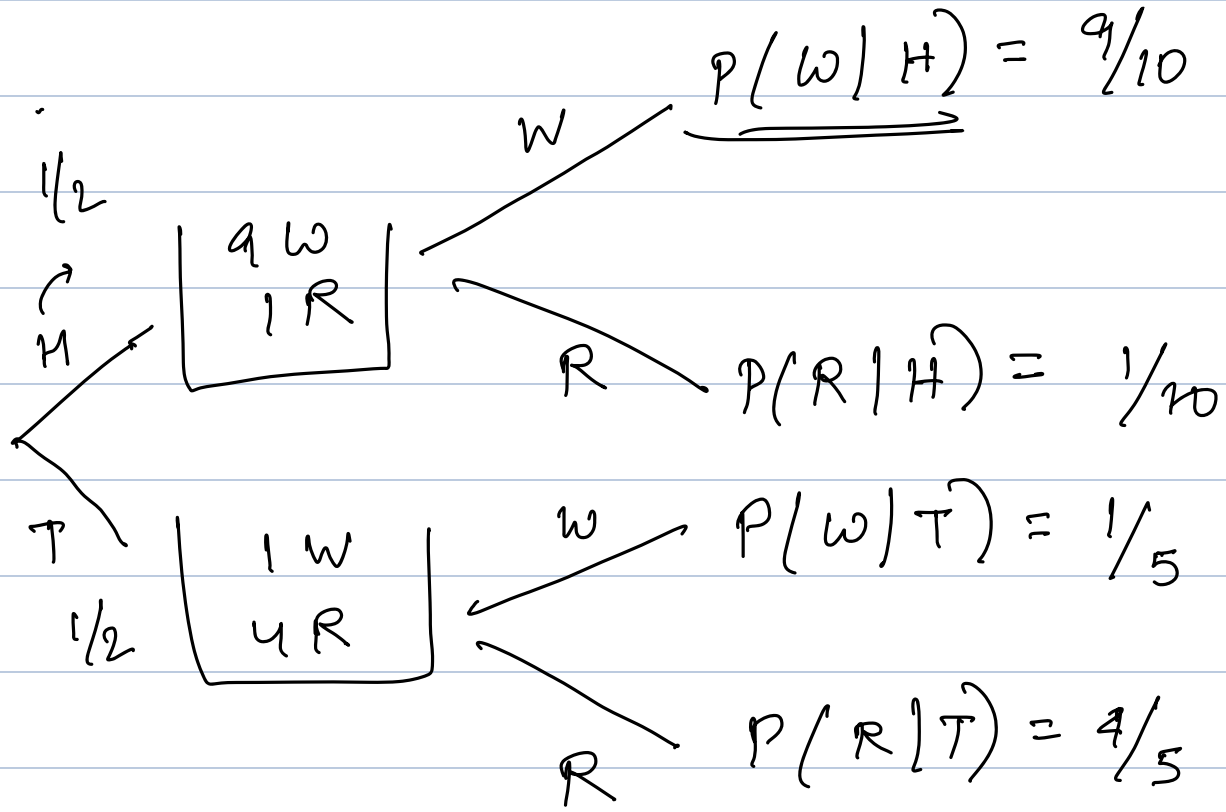
(ii) We then pick a ball at random from the chosen urn

$P(W)$ → W = "pick a white ball"

H = coin shows head

R = "pick a red ball"

T = coin shows tails



$$P(W) = P(W|H)P(H) + P(W|T)P(T)$$

$$= \frac{9}{10} \cdot \frac{1}{2} + \frac{1}{5} \cdot \frac{1}{2}$$

$$= 11/20$$

Q. Assume that at the end, we have picked a white ball. What are the chances that we tossed heads in the first place?

$P(H|W)$?

$$P(H|W) = \frac{P(W|H) \cdot P(H)}{P(W)}$$

$$= 9/11$$

Eg. 3 urns

#1 5W, 1R

#2 0W, 7R

#3 3W, 3R

Game: Roll a fair die

$B_1 = \{1,2,3\}$ choose urn 1

$B_2 = \{4\}$ " " 2

$B_3 = \{5,6\}$ " " 3

then pick a random ball.

$P(R) = 5/12$

$P(B_2|R) = ?\ 2/5$

E.g. Suppose that we are testing for a disease

Afflicts 1% of population

Suppose the patient tests positive.

Suppose the test is 95% accurate

D = "patient has disease"

T = " " has tested +ve"

$$P(D \mid T) = ?$$

$$= \frac{\overset{0.95}{P(T \mid D)}\, \overset{0.01}{P(D)}}{P(T)}$$

$$P(T) = \underbrace{P(T \mid D)}_{\text{[illegible]}}\, P(D) + P(T \mid D^c)\, P(D^c)$$

$$= 0.95,\ 0.01$$

$$+ \; 0.05 \cdot 0.99$$

$$= 0.059$$

$$P(D|T) = \frac{0.95 \cdot 0.01}{0.059}$$

$$\simeq 0.16$$

Conditional Probabilities are also probabilities

(i) $P(S|E) = 1$,

(ii) $P(\phi|E) = 0$

Prove this

(iii) $A_1, A_2 \ldots\ldots$ are disjoint

$$P\left(\bigcup_{j=1}^{\infty} A_j \,\middle|\, E\right) = \sum_{j=1}^{\infty} P(A_j|E)$$

$$P(A^c|E) = 1 - P(A|E)$$

$$P(A \cup B \mid E) = P(A \mid E) + P(B \mid E) - P(A \cap B \mid E)$$

Q. Consider an election with 2 candidates, Candidate A & " B

For a randomly selected voter,

let A : event that they voted for candidate A

W : event that they are willing to participate in the exit poll.

Suppose : $P(W \mid A) = 0.7$

$P(W \mid A^c) = 0.3$

In the exit poll, 60% of the respondents say that they voted for A.

(Assume are honest)

Find $P(A) =$

Given $P(A|\omega) = 0.6$

$$P(A|\omega) = \frac{P(\omega|A)\,P(A)}{P(\omega)} \quad \text{—} \; \star$$

$P(A) = x$

$P(\omega)$

$$= P(\omega|A)\,P(A) + P(\omega|A^c)\,P(A^c)$$

$$= P(\omega|A)\,P(A) + P(\omega|A^c)\,(1 - P(A))$$

Substitute in $\star$

$$P(A|\omega) = \frac{P(\omega|A)\cdot x}{P(\omega|A)\,x + P(\omega|A^c)\,(1-x)}$$

$$x = 9/23 \approx 0.391$$

Q. Company A has just developed a diagnostic test for a certain disease. The disease afflicts 1% of the population.
The probability of someone testing +ve, given that they have the disease = 0.95

The probability of someone testing -ve, given that they don't have the disease = 0.95

Company B offers a competing test for the disease.
Company B claims that their test is faster & less expensive to perform than Company A's test, is less painful, & yet has a higher $\underbrace{\text{overall success rate}}_{\text{probability that a random person gets diagnosed correctly}}$,

(a) Company B's test: No matter who the patient is, diagnose that they do not have the disease.
Check whether Company B's claim about overall success rates is true.

Sol^n: Company B's test: Diagnose "no disease" always

Given: $P(D) = 0.01$ D: having disease

$P(D^c) = 0.99$

$P(+ve \mid D) = 0.95$

$P(-ve \mid D^c) = 0.95$

$P(-ve \mid D) = 0.05$

Overall success rate = probability diagnosis is correct

From Company B's diagnosis

No disease $\Rightarrow$ Diagnosis is correct

Disease $\Rightarrow$ " " incorrect

Therefore Success rate for company B:

$$P(D^c) = 0.99$$

Success rate for company A:

$$P(+ve \cap D) + P(-ve \cap D^c)$$

$$= P(+ve \mid D)\,P(D) + P(-ve \mid D^c)\,P(D^c)$$

$$= 0.95 \cdot 0.01 + 0.95 \cdot 0.99$$

$$= 0.95$$

Success rate for company B > success rate for company A